

TANISHPREET SINGH SIDHU

(aka Tanish Sidhu, pronounced: Taniš)

B.Techtanishpreet.official@gmail.com | [linkedin.com/in/tanishsidhu](https://www.linkedin.com/in/tanishsidhu)

Valid Serbian Work Permit



PROFESSIONAL SUMMARY

Software Engineer focused on applied AI and agentic architecture, bringing 7 years of enterprise software experience building and managing high-volume product implementations. I specialize in integrating language models with business applications, and I design agentic workflows that solve complex operational problems, leveraging my engineering background and combining my robust C#, Python and SQL foundation with practical AI techniques. Based in Belgrade, Serbia, with full work authorization and no visa sponsorship required.

SKILLS & COMPETENCIES

Languages and Databases: C#, Python, SQL (T-SQL, PostgreSQL), Entity Framework, LINQ, HTML, JavaScript, TypeScript

AI & Applied Engineering: Local LLM Fine-Tuning, RAG Architectures, Agentic Workflows, LangGraph, LLM API Integrations (Anthropic, OpenAI, DeepSeek), Prompt Engineering, Claude Code, Cursor, Codex, OpenClaw, Hermes AI, OpenCode

Cloud & DevOps: Git, Azure (Functions, CloudOps), Supabase, Vercel, CI/CD Pipelines, Postman

Data Analysis, Visualization, & Collaboration: Microsoft Office Suite, Power Automate, JIRA, ServiceNow, Confluence, Slack

PROFESSIONAL EXPERIENCE

ODESSA (FinTech, Asset Finance Product)

Technical Lead – (April 2025 – May 2026)

Staff Engineer – (April 2024 – March 2025)

Senior Software Engineer – (April 2022 – March 2024)

Software Engineer – (June 2020 – March 2022)

Intern – (June 2019 – May 2020)

June 2019 – May 2026

Belgrade, Serbia

Bengaluru, Karnataka, India

- Led a globally distributed engineering and functional team across **U.S, India and Serbia**, delivering complex FinTech implementations for enterprise clients with over **\$5 billion in revenue**. Alongside core technical delivery, championed internal AI initiatives by exploring integrations with external AI vendors to build intelligent features for our clients.
- Drove the adoption of AI developer tools across the engineering team, integrating **Claude Code**, **Cursor**, and **Codex** directly into daily workflows. Utilized these models to accelerate code generation, assist in deep debugging, and establish practical standards for **AI assisted development**.
- Owned core backend development and deep system diagnostics for high stakes enterprise environments, supporting global clients including **Hewlett Packard Enterprise, Canon, GE Capital, Volvo, and TD Bank**. Engineered robust solutions using **C#** and **SQL** to resolve complex transactional discrepancies for these clients, ensuring the strict data integrity required for accurate financial processing and downstream integrations.
- Engineered high performance database solutions using **SQL query optimization** techniques, successfully improving system **performance by up to 70%**. Refactored complex stored procedures to ensure the rapid, scalable data retrieval necessary for high volume enterprise platforms.
- Handled **API integrations and troubleshooting** to seamlessly connect enterprise client workflows, utilizing **Postman** to inspect raw requests and responses and validate payload structures. Debugged failing endpoints and data mapping errors, directly assisting client teams in successfully embedding Odessa APIs into their core business workflows.
- Managed **cloud deployments** and both internal and production releases across **Azure** environments. Handled complex versioned releases, including preparing manual release content and executing controlled deployments when automation was not feasible, ensuring strict rollback readiness and minimal downtime.
- Maintained engineering standards through rigorous peer **code reviews** and comprehensive **unit testing**. Ensured the delivery of clean, maintainable code that minimized post-release bugs and reduced overall rework.
- Supported full-cycle feature development and **product upgrades**, from initial code integration through **performance testing**. Led UAT sessions with business stakeholders to ensure technical deliverables met exact business requirements before release.

SELECT PROJECTS

- **Payload-Gen | Locally Fine-Tuned LLM & RAG Architecture** – Architected a natural language to API payload generator by combining a retrieval augmented generation pipeline for schema retrieval with a locally fine-tuned Llama 3B model. Designed deterministic evaluation methods to validate output, achieving a 95% valid JSON payload generation rate on unseen enterprise API specifications. Demonstrated a secure, low latency, on-premise alternative to frontier models for high volume enterprise integrations. (github.com/tanishsidhu/payloadgen)
- **Support-Agent | Agentic Workflow** – Developed a support agent prototype utilizing an agentic loop architecture and DeepSeek to handle enterprise ticket classification. Engineered a retrieval system to pull resolution evidence from historical documentation, enabling the agent to autonomously draft highly accurate, cited responses. (github.com/tanishsidhu/supportagent)
- **Pokrit | Full Stack AI SaaS** – Built and deployed an end to end AI SaaS platform that autonomously scrapes job boards and utilizes language models to match live listings against user resumes. Engineered the backend using C# .NET 8, and Supabase (PostgreSQL), with a Next.js frontend, generating daily digest emails with fit scores and automated cover letters. (pokrit.vercel.app)
- **PeakOrLow | LLM API Integration & Web Application** – A full stack market briefing web application utilizing C# and ASP.NET Core. Integrated the Anthropic Claude API to process data and generate automated intelligence summaries. (peakorlow.azurewebsites.net)

AWARDS & CERTIFICATIONS

- **Building with the Claude API** – *Certified by Anthropic*
- **Claude Code in Action** – *Certified by Anthropic*
- **SQL** – *HackerRank Skill Certification*
- **Python Professional** – *OpenEDG Python Institute & LinkedIn Learning*
- **LangGraph Essentials** – *Certified by LangChain Academy*
- **Foundations of Project Management** – *Certified by Google and Coursera*
- **Artificial Intelligence Essentials** – *LinkedIn Learning Program*
- **Data Structures & Java** – *Certified by Coding Ninjas*

EDUCATION

CHITKARA UNIVERSITY - (CGPA: 9.17)

Bachelor of Technology – BTech, Computer Science

2016 – 2020

Rajpura, Punjab, India

GIAN JYOTI GLOBAL SCHOOL | THE BRITISH SCHOOL - (CGPA: 10.00)

High School – 10th to 12th Standard

2014 – 2016

Chandigarh, India

LANGUAGES

- **English** – *Full Professional Proficiency*
- **Hindi** – *Native Proficiency*
- **Punjabi** – *Native Proficiency*
- **Serbian** – *Beginner*